Semester	B.E. Semester VII – Computer Engineering
Subject	Scalable ML & BDA Lab
Subject In-charge	Prof. Shweta Jaiswar

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Grade and Subject Teacher's Signature	

Experiment Number	05	
Experiment Title	**Hyperparameter Tuning for Machine Learning Models**	
Resources / Apparatus Required	Hardware:	Software:
Description	Optimize the performance of a machine learning model using Optuna. Train a Decision Tree or Random Forest classifier, automatically search for the best hyperparameters, compare model performance before and after optimization, and analyze the impact of hyperparameter tuning on prediction accuracy.	
Observation	The Random Forest classifier was first trained using default hyperparameters and its performance was evaluated using accuracy, F1-score, precision, and recall. Optuna was then used to automatically search for optimal values of parameters such as the number of trees, maximum tree depth, minimum instances per node, maximum bins, and feature subset strategy. The optimized Random Forest was trained using the best hyperparameters obtained from the Optuna study and was evaluated on the same test dataset. The comparison showed that hyperparameter tuning improved the overall predictive performance of the model compared with the default Random Forest.	
Program	HyperParameter_Tuning_Exp_SML	

Output

```
root
 |-- customerID: string (nullable = true)
 |-- gender: string (nullable = true)
 |-- SeniorCitizen: integer (nullable = true)
 |-- Partner: string (nullable = true)
 |-- Dependents: string (nullable = true)
 |-- tenure: integer (nullable = true)
 |-- PhoneService: string (nullable = true)
 |-- MultipleLines: string (nullable = true)
 |-- InternetService: string (nullable = true)
 |-- OnlineSecurity: string (nullable = true)
 |-- OnlineBackup: string (nullable = true)
 |-- DeviceProtection: string (nullable = true)
 |-- TechSupport: string (nullable = true)
 |-- StreamingTV: string (nullable = true)
 |-- StreamingMovies: string (nullable = true)
 |-- Contract: string (nullable = true)
 |-- PaperlessBilling: string (nullable = true)
 |-- PaymentMethod: string (nullable = true)
 |-- MonthlyCharges: double (nullable = true)
 |-- TotalCharges: string (nullable = true)
 |-- Churn: string (nullable = true)
```

Data set snapshot

```
+-----+-----+
|Churn|count|
+-----+-----+
|   No| 5163|
|  Yes| 1869|
+-----+-----+
```

Data count by churn label

```
Categorical columns: 15
Numerical columns: 4
```

```
Training rows: 5690
Testing rows: 1342
```

Columns Types and separation of training and testing data

```
Training data:
+--------------------+-----+
|            features|label|
+--------------------+-----+
|(45,[0,3,5,6,8,12...|  0.0|
|(45,[1,2,4,6,9,12...|  0.0|
|(45,[1,3,4,6,8,11...|  1.0|
|(45,[0,3,4,6,8,11...|  1.0|
|(45,[0,2,5,6,8,12...|  0.0|
+--------------------+-----+
only showing top 5 rows

Testing data:
+--------------------+-----+
|            features|label|
+--------------------+-----+
|(45,[1,2,4,6,8,11...|  1.0|
|(45,[0,3,4,6,8,11...|  0.0|
|(45,[0,2,4,6,8,12...|  0.0|
|(45,[0,2,4,6,9,11...|  0.0|
|(45,[1,2,4,7,10,1...|  1.0|
+--------------------+-----+
only showing top 5 rows
```

Vector Assembly of Training and Testing data

```
===== DEFAULT RANDOM FOREST =====
Accuracy : 0.7951
F1 Score : 0.7722
Precision: 0.7826
Recall   : 0.7951
```

Random Forest Outputs

```
Optuna training rows: 4598
Validation rows: 1092
```

```
===== BEST OPTUNA RESULT =====
Best Validation Accuracy:
0.8086

Best Hyperparameters:
numTrees: 40
maxDepth: 15
minInstancesPerNode: 3
maxBins: 32
featureSubsetStrategy: log2
```

Rows used for optuna training and it's result

```
===== OPTIMIZED RANDOM FOREST =====
Accuracy : 0.7973
F1 Score : 0.7894
Precision: 0.7870
Recall   : 0.7973
```

Optimized Random Forest

	Metric	Default Random Forest	Optimized Random Forest	Improvement
0	Accuracy	0.795082	0.797317	0.002235
1	F1 Score	0.772246	0.789408	0.017162
2	Precision	0.782581	0.786998	0.004417
3	Recall	0.795082	0.797317	0.002235

Benchmarking Metrics for evaluation of both Random Forest Models

Conclusion:

The experiment successfully demonstrated that Optuna can be used to automate hyperparameter optimization for a Random Forest classifier. The optimized model performed better than the default model, showing that appropriate selection of hyperparameters can improve prediction accuracy and other evaluation metrics. Therefore, hyperparameter tuning is an effective technique for improving machine learning model performance and can help obtain a more suitable model for the given dataset.